

Python DA Assignment 2 – Data Visualization

Project Summary

Project Overview

This project focuses on analyzing and visualizing taxi trip data using **Python, Pandas, Matplotlib, and Seaborn**. The Seaborn built-in **Taxis dataset** was used to understand patterns related to taxi fares, trip distances, payment methods, tips, and pickup locations.

Objective

The main objective of this project was to perform **data cleaning, exploratory data analysis, and data visualization**. The project demonstrates how raw taxi data can be transformed into meaningful visual insights that support better understanding of trip and customer behavior.

Data Cleaning

The dataset was first inspected for missing values. Missing numerical values were handled using an appropriate statistical method such as the **median**, while categorical missing values were filled using the **mode**. The pickup column was converted to datetime format to support time-based analysis.

Visualizations Created

Matplotlib / Pandas

- **Line Chart:** Analyzed fare changes over pickup time.
- **Bar Chart:** Compared total fare across pickup boroughs.
- **Pie Chart:** Displayed the distribution of trips by payment method.
- **Histogram:** Analyzed the distribution of taxi trip distances.
- **Box Plot:** Compared tip distributions across pickup boroughs.

Seaborn

- **Count Plot:** Compared the number of trips across pickup boroughs.
- **Scatter Plot:** Examined the relationship between distance and fare, categorized by pickup borough.
- **Heatmap:** Analyzed correlations between distance, fare, tip, tolls, and total.
- **Pair Plot:** Explored pairwise relationships among distance, fare, tip, and total across pickup zones.

- **Violin Plot:** Compared fare distributions across different payment methods.

Key Insights

The visualizations help show the relationship between **trip distance and fare**, differences in trip activity across pickup boroughs, variation in tip amounts, and differences between payment methods. The correlation heatmap provides a clear view of how important numerical variables are related, while the pair plot gives a broader view of relationships among key taxi trip measures.

Conclusion

This project demonstrates a complete basic data-analysis workflow using Python. By combining **Pandas for data preparation** with **Matplotlib and Seaborn for visualization**, the taxi dataset was transformed into understandable graphical insights. The assignment strengthened practical skills in missing-value handling, data manipulation, exploratory data analysis, and data visualization.